

$$N \propto \frac{Eb}{\phi}$$

$$N_{se} \propto \frac{V/2}{I}$$

$$N_{se} \propto \frac{V}{2I}$$

$$N_{sh} = 4 N_{se}$$

$$N \propto \frac{Eb}{\phi}$$

$$N_{sh} \propto \frac{V}{I/2}$$

$$N_{sh} \propto \frac{2V}{I}$$

$$\begin{matrix} Eb \propto V \\ \phi \propto I \end{matrix}$$

$$T \propto \phi I_a$$

$$T_{se} \propto I^2$$

$$T_{sh} = \frac{T_{se}}{4}$$

$$T \propto \phi I_a$$

$$T_{sh} \propto \frac{I}{2} \cdot \frac{I}{2}$$

$$T_{sh} \propto I^2/4$$

* STARTERS →

* If a dc motor started directly with rated vol. across the terminal it will draw excessively high current.

* If the motor starts quickly this will be inrush current which may not damage the motor.

* Large motors which have small acceleration time comparatively draw huge current which produce vol. dip in the supply where it is connected & also damages the commutator & brush. Therefore starting resistance should be inserted in series with arm, which should be cut down in steps.

* A starter insure this starting resistance to limit the starting current to a desired value.

* There are 3 more protective schemes :-

① No volt release

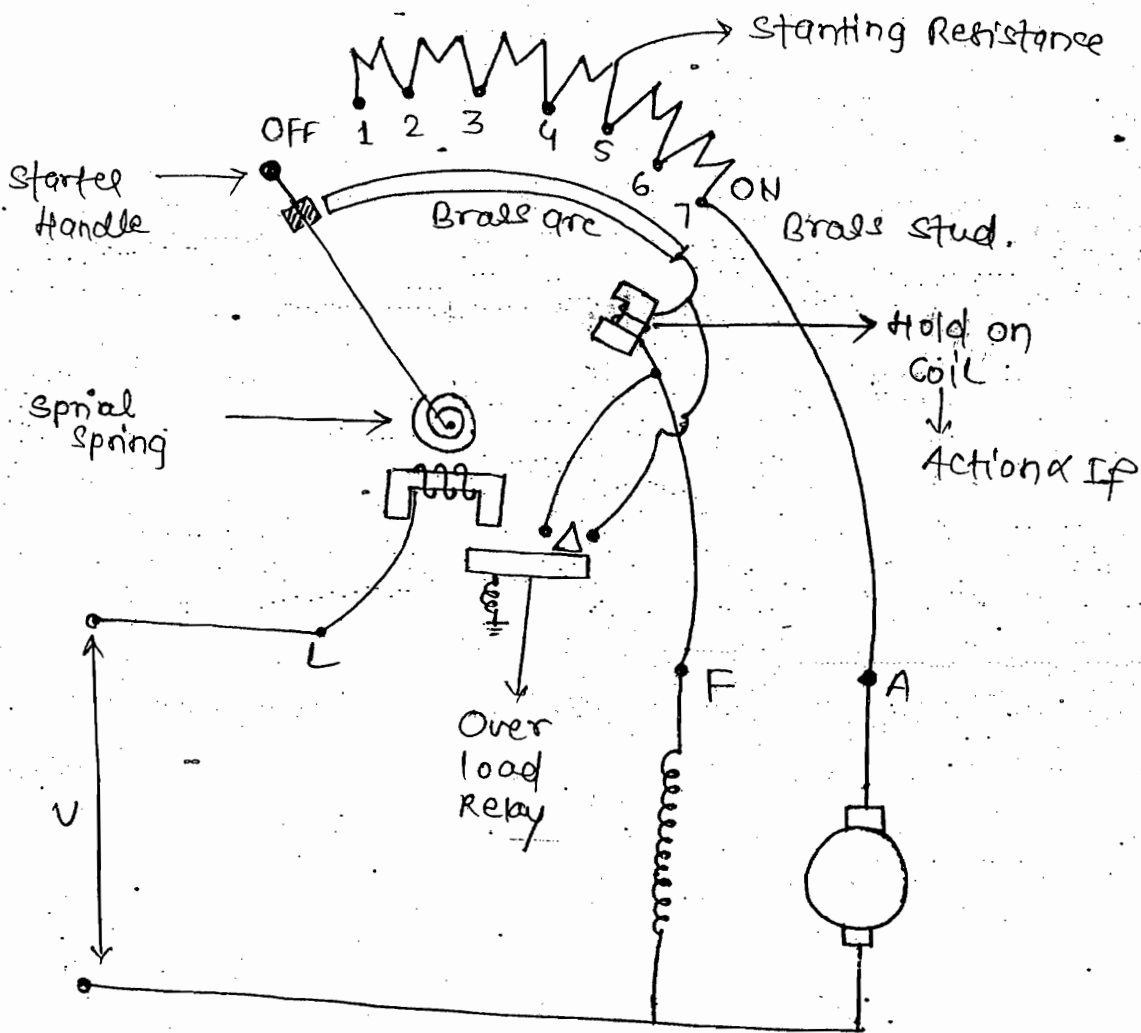
③ Field failure prevention

② Over load release

* There are 3 types of starters:-

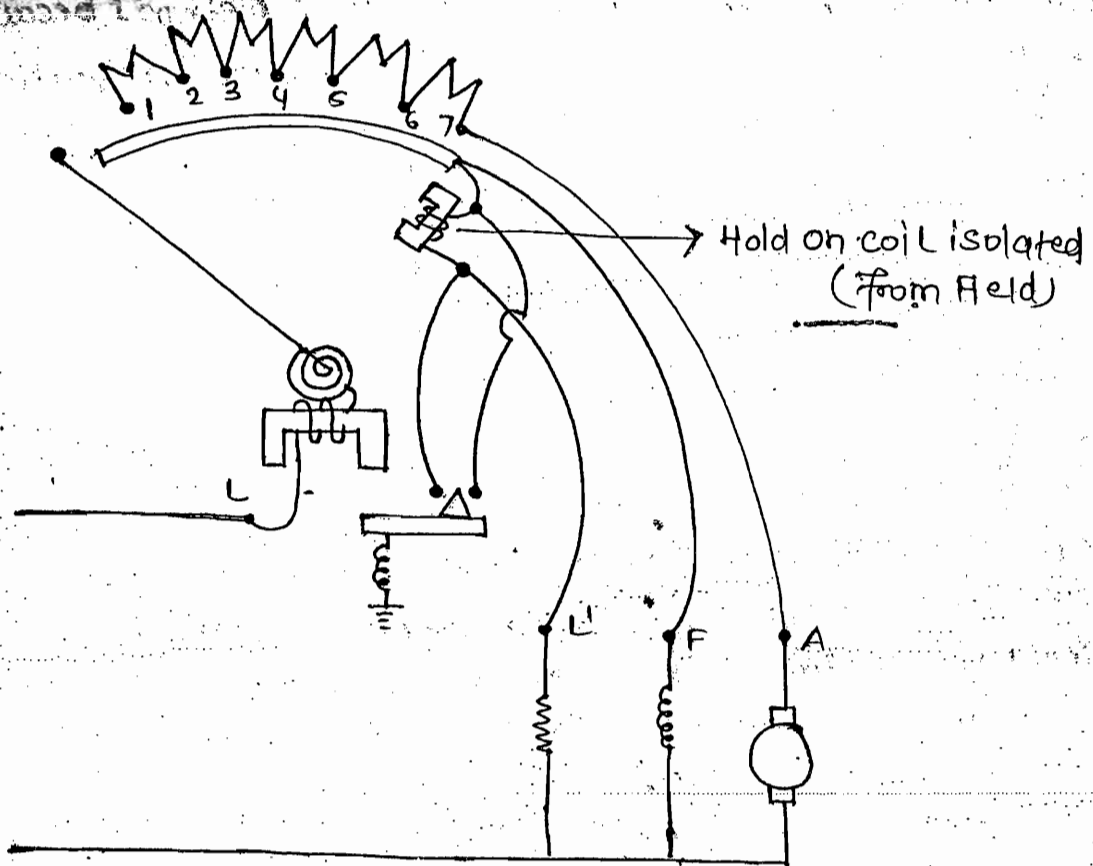
- 3 point starter } Shunt + Compound
- 4 point starter }
- 2 point starter → Series

* 3 point starter →

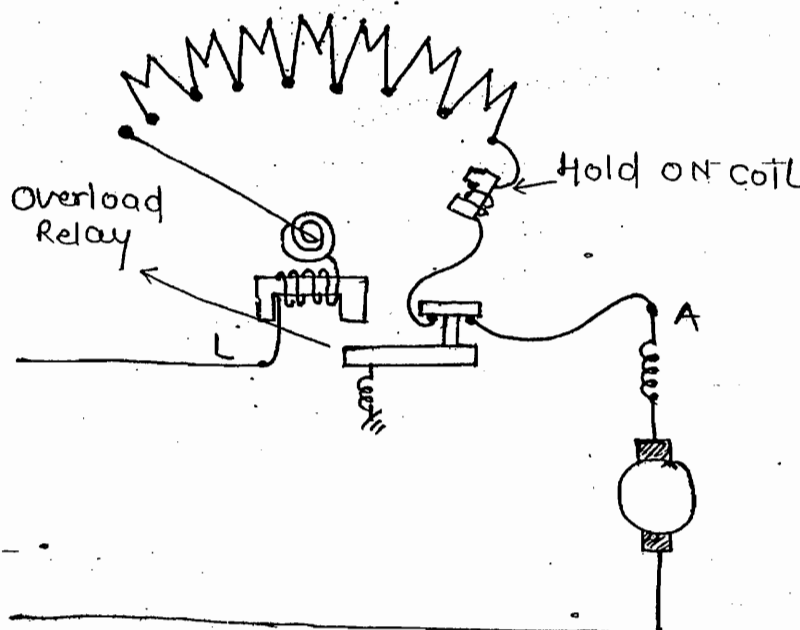


* 3 point starter →

Sagar Sen
887145336



* 2 point starter →



3 point starters are not suitable for shunt & compound motor when they are subjected to Fwsc method because the Hold on coil magnetic action is directly proportional to its field current.

For such applⁿ 4-point starter should be used.

Its Hold on coil is isolated from field wdg even though field current decreased. it won't effect Hold on coil action but there is no field failure prevention.

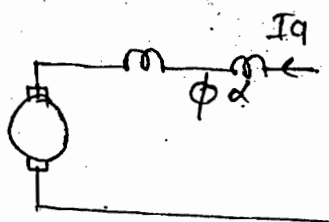
2 point starters are specially used for series motor only which contains all the 4-schemes.

Apart from them it also protect the motor at any dangerous speeds (racing condⁿ) when the load across its shaft is suddenly disconnected because the Hold on coil is in series with field wdg & arm.

A 3 Pt starter can be used to start a series motor but with a slight modification.

Its F terminal should be closed through -ve to offer a closed path for current to flow in the Hold on coil.

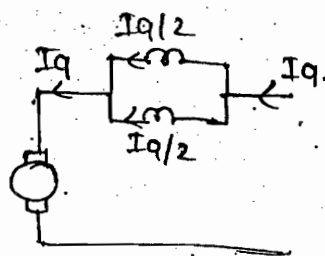
(4)
c)



$$T \propto \phi \cdot I_q$$

$$T \propto I_q \cdot I_q$$

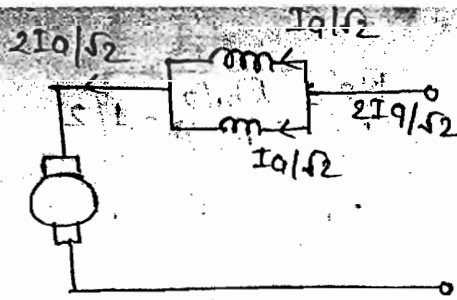
$$T \propto I_q^2$$



$$T \propto \phi I_q$$

$$T \propto \left(\frac{I_q}{2} \right) (I_q)$$

$$T \propto \frac{I_q^2}{2}$$



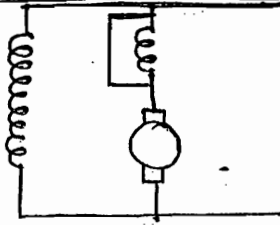
$$T \propto \phi I_a$$

$$T \propto \frac{I_a}{\sqrt{2}} \times \frac{2I_a}{\sqrt{2}}$$

$$T \propto I_a^2$$

If flux is $\phi/\sqrt{2}$ then speed $N\sqrt{2}$.

(31/1) Rated torque prob. $\rightarrow$



For cumulatively

$\phi_{sh} + \phi_{se}$ $\phi \uparrow$
series gets SC

ϕ_{sh} $\phi \downarrow$

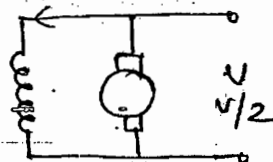
$$\uparrow N \propto \frac{E_b}{\phi} \downarrow$$

$$T \propto \phi I_a \uparrow$$

Flux reduce & to maintain constant torque motor will draw current high.

(34/7) Rated power $\rightarrow$

$$I_{ch} = V/R_{sh}$$



$$I_{ch} = \frac{V}{2R_{sh}}$$

$$\phi \rightarrow \phi_{1/2}$$

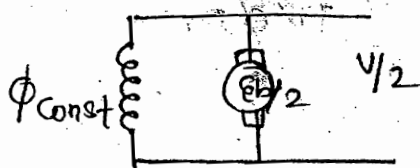
$$\frac{N_2}{N_1} = \frac{E_{b1/2}}{E_{b1}} \times \frac{\phi_1}{\phi_{1/2}}$$

$$N_2 = N_1 \quad \text{Speed } \downarrow \text{ pu}$$

$$P = E_b I_a$$

$E_b = E_b/2$ then $I_a = 2I_a$ then power constant

38/8



$$\frac{N_2}{N_1} = \frac{E_{b1}/2}{E_{b1}} = 1/2$$

$$N_2 = 0.5 N_1 \text{ PU}$$

$$P = E_b I_a$$

$$I_a = 2 \text{ PU}$$

DATE 10/07/14

* BRAKING → * It is done to intentionally stop the motor or to control its speed.

* There are 2 basic brakings

(i) Mech. braking → The KE of the rotating parts is dissipated in brake which includes noise, wear & tear, high maintenance repair & the braking is not smooth.

(ii) Ele. braking → Isolating the motor from supply is basic electric braking but the motor doesn't stop.

In order to stop the motor at the required instant quickly additional braking methods are used.

(a) Dynamic/Rheostatic braking

(b) Plugging

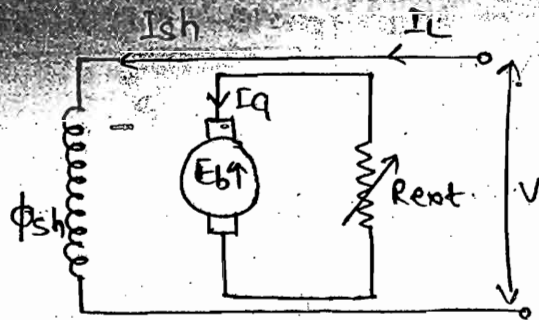
(c) Regenerative braking.

(a) Dynamic/Rheostatic braking → * The basic principle involved in ele. braking is to

develop a -ve torque in a running motor with the reversal of arm. current.

* In this braking the arm. of a running motor is disconnected from supply leaving the field connected to supply,

* At the instant of braking the induced back emf will circulate a current in the opposite dirⁿ to reverse the torque & motor stops quickly.

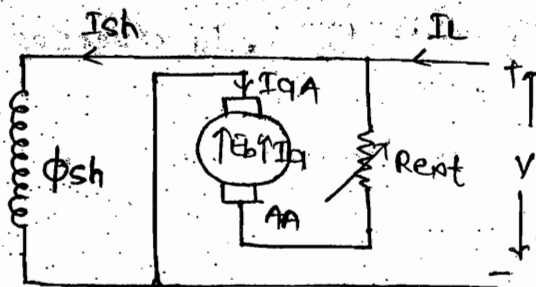


$$I_b = \frac{E_b}{R_a + R_{ext}}$$

(2) Plugging → * Reversing the arm. terminals or arm. polarity only which will directly reverse the arm. current & torque.

* Due to which the motor want to run in opposite dirn.

* Consequently it come near 0 speed where the mech. braking will be applied otherwise the motor continues to run in opposite dirn.



$$I_a = \frac{V + E_b}{R_a + R_{ext}}$$

(43)
9

$$E_b = V - I_a R_a$$

$$I_L = 15A, R_{sh} = 80\Omega, V = 240V$$

$$I_{sh} = \frac{V}{R_{sh}} = \frac{240}{80} = 3A$$

$$I_a = I_L - I_{sh}$$

$$I_a = 12A$$

$$E_b = V - I_a R_a$$

$$= 240 - 12(0.5)$$

$$= 234$$

$$V + E_b = 234 + 240 = 474V$$

44
9

$$I_a = \frac{E_b + V}{R_a + R_{ext}}$$

$$-I_a (1.25)(12) = \frac{240 + 234}{0.5 + R_{ext}}$$

$$R_{ext} = 31.1 \Omega$$

(3) Regenerative Braking → * This is not intentional & it won't stop the motor. It occurs naturally.

* It occurs naturally due to the inherent property of motor when it is subjected to over-hauling load condn like a train moving down a gradient a crane lowering its load etc.

* Due to increased speed if $E_b > V$, I_a reverses & the torque reverses & the speed is reduced.

* Consequently the motor regains its original mode ($E_b < V$)

* During braking it is acting as gen simultaneously.

* In series motor in order to achieve this braking its field wdg need to be separately excited.

For series motor

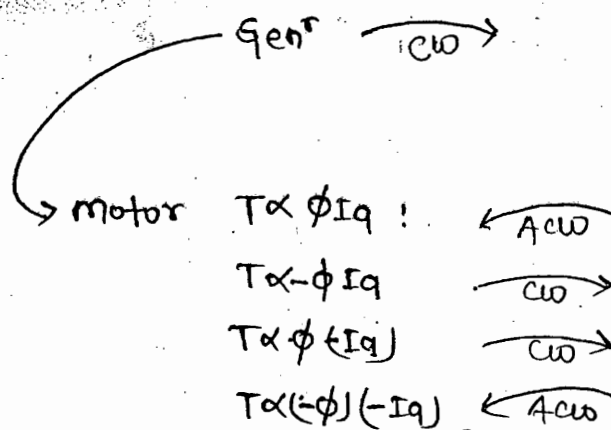
$$T = k\phi (-I_a)$$

↓
+ve

* So in series motor we brake the arm. & a series field.

* This are so advantage in the mountain Railway.

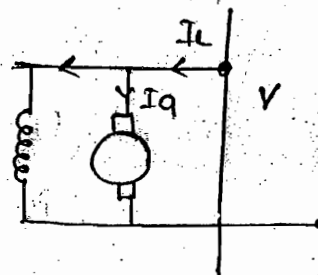
* Gen^r & motor both operating at motor →



⊗ A shunt gen^r supply-ing power across bus bar when the prime mover fails it will act as

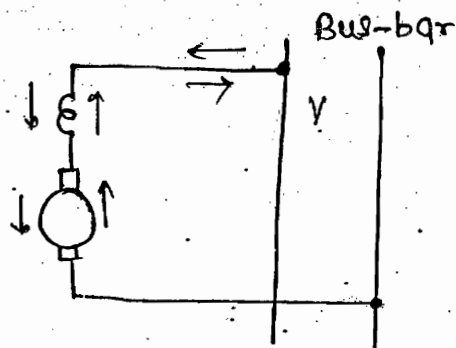
- (a.) shunt motor running at same dirⁿ.
- (b.) Shunt motor running in opposite dirⁿ.
- (c.) It will stop.

Ans. (a.) (Only I_a dirⁿ change)

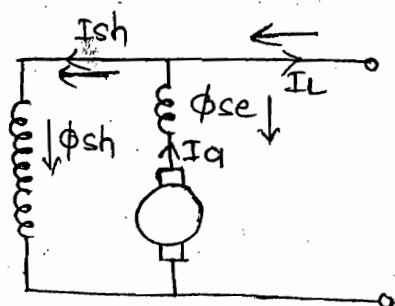


⊗ Series gen^r supplying power to a bus bar, when its prime mover fails it will act as

- (a.) Series motor running at same dirⁿ
- (b.) Series motor running at opposite dirⁿ



Ans → (b) (Both ϕ & I_a change)



So by reversing the supply or failure of prime mover there is a reversal of I_a & cumulatively compound becomes diff compound because of the same dirn in the change in dirn.

(21/5)

$$\phi_1 \rightarrow \phi_1, \phi_2 \rightarrow 1.1\phi_1$$

$$E_{b1} = V - (I_a R_a) \text{ (drop)}$$

$$E_{b2} = 0.95 E_{b1}$$

$$\frac{N_2}{N_1} = \frac{E_{b2}}{E_{b1}} \times \frac{\phi_1}{\phi_2} =$$

$$\frac{N_2}{1000} = \frac{0.95 E_{b1}}{E_{b1}} \times \frac{\phi_1}{1.1\phi_1}$$

$$N_2 = 863 \text{ rpm}$$

$$N \propto \frac{E_b}{\phi}$$

$$N \propto E_b \propto V - I_a R_a$$

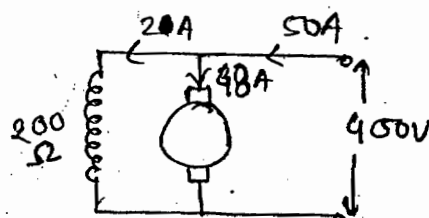
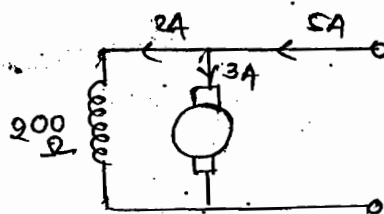
$$E_b \text{ at NL} = 400 - 5 \times (0.5 + 200) = 398.5 \text{ V}$$

$$E_b \text{ at FL} = 400 - 48(0.5)$$

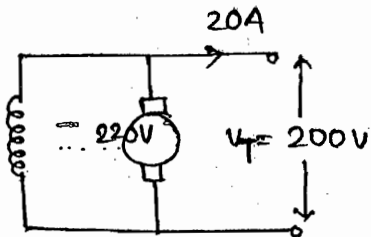
$$= 376 \text{ V}$$

$$\frac{N_1}{N_2} = \frac{E_{b1}}{E_{b2}}$$

$$\frac{998.9}{376} = \frac{N_2}{111} = \frac{376}{398.5} = 0.94$$



const: $E_b \propto \phi N \uparrow$ $\downarrow T \propto \phi I_a$



$$\phi_2 = 1.1 \phi_1$$

$$E_g = V + I_a R_a$$

$$= 200 + 20(0.2)$$

$$E_g = 204V$$

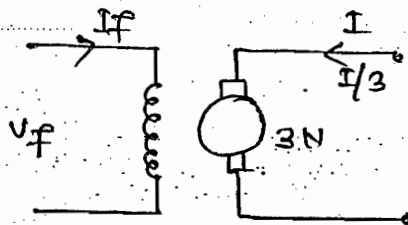
$$\frac{N_2}{N_1} = \frac{196}{204} \times \frac{\phi_1}{\phi_1 \cdot 1.1}$$

$$E_b = V - I_a R_a$$

$$= 200 - 20(0.2)$$

$$= 196V$$

$$\boxed{\frac{N_2}{N_1} = 0.87}$$



$$P = 3 E_b \cdot I_a / 3$$

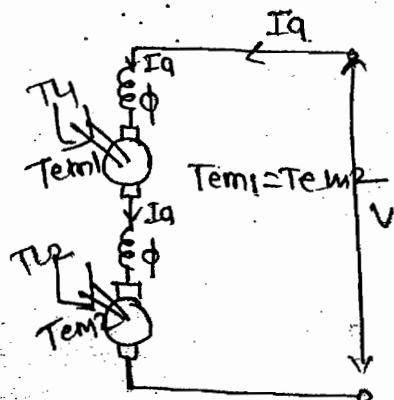
Constant power

$$T \propto I_a \phi$$

$$N \propto \frac{E_b}{\phi}$$

$$T \propto \phi I_a$$

$$T \uparrow N \uparrow$$



$$P = \frac{2\pi NT}{60}$$

$$P \propto NT$$

Because same T

$$P \propto N, P_1 \propto N_1, P_2 \propto N_2$$

$$T_1 = T_2, T_{em1} = T_{em2}$$

$$\boxed{1:1}$$

If in ques given two diff motor then ans: $N_1: N_2$

T_{em} - Electromagnetic torque

T_L - load torque